RP5: BI-RADS-GROUNDED RETRIEVAL-AUGMENTED GENERATION ASSISTANT FOR BREAST IMAGING REPORTING

PROJECT REPORT [Phase-II]

Submitted in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY
Department of Computer Science and Engineering

Submitted to

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL (M.P.)

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October 2026

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY
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CERTIFICATE

This is to certify that the work embodied in this project report entitled **"RP5: BI-RADS-Grounded Retrieval-Augmented Generation Assistant for Breast Imaging Reporting"** has been satisfactorily completed by **Akash Singh Bhadoria ([Scholar No. 1])** and **Alok Yadav ([Scholar No. 2])**. It is a bonafide piece of work, carried out under our guidance in the Department of Computer Science and Engineering, Indian Institute of Information Technology, Bhopal for the partial fulfillment of the award of the degree of **Bachelor of Technology** in Computer Science and Engineering during the academic session 2025–2026.

The results embodied in this report have been thoroughly verified and have not been submitted to any other University or Institute for the award of any degree or diploma.

Date: 13th October 2026

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DECLARATION

We hereby declare that the major project report entitled **"RP5: BI-RADS-Grounded Retrieval-Augmented Generation Assistant for Breast Imaging Reporting"** presented here is the partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**.

It is an authentic documentation of our original work carried out under the able guidance of **[Supervisor Name]**, Department of Computer Science and Engineering. The work has been carried out entirely at the **Indian Institute of Information Technology, Bhopal**. The project work presented has not been submitted in part or whole for the award of any degree or professional diploma in any other institute or organization.

We, with this, declare that the facts, algorithmic workflows, code implementations, and experimental results mentioned above are true to the best of our knowledge. In case of any unlikely discrepancy that may occur, we will be the ones to take responsibility.

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ACKNOWLEDGEMENT

We, **Akash Singh Bhadoria** and **Alok Yadav**, would like to express our sincere gratitude to everyone who supported and guided us throughout the successful completion of our project titled **“RP5: BI-RADS-Grounded Retrieval-Augmented Generation Assistant for Breast Imaging Reporting”**.

First and foremost, we would like to extend our heartfelt thanks to our esteemed **Director, [Director's Name]**, Indian Institute of Information Technology Bhopal, for providing us with the necessary computing resources, research infrastructure, and scholarly environment for conducting our project. His encouragement and vision played a pivotal role in helping us accomplish our research and engineering goals.

We are deeply grateful to our **Project Guide, [Supervisor Name]**, [Supervisor Designation], Department of Computer Science and Engineering, for his constant guidance, invaluable suggestions, rigorous technical reviews, and expert advice throughout the project. His deep insights into artificial intelligence, clinical natural language processing, and system evaluation helped us enhance the quality of our work, and his steadfast support kept us motivated during difficult debugging and experimental phases.

Our sincere thanks to the **Department Coordinator, [Coordinator's Name]**, for the valuable administrative support and encouragement provided during the completion and evaluation of the project.

We would also like to express our gratitude to all the faculty members of the **Department of Computer Science and Engineering** for their valuable guidance, insightful feedback during project presentations, and support in clarifying technical concepts throughout the smooth execution of our project.

A special thanks to our **friends and peers** for their continuous support, valuable suggestions, collaborative brainstorming, and help during various stages of data structuring and adversarial scenario design. Their encouragement motivated us to maintain academic rigor and enthusiasm.

Last but certainly not least, we express our deepest gratitude to our **parents and family members** for their unconditional love, patience, sacrifices, and continuous moral support throughout the course of our undergraduate studies. Their faith in our capabilities was the primary driving force behind our efforts.

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AREA OF WORK

Our project focuses primarily on the specialized interdisciplinary domains of **CLINICAL NATURAL LANGUAGE PROCESSING (NLP)**, **RETRIEVAL-AUGMENTED GENERATION (RAG)**, and **SAFEGUARDED CLINICAL DECISION SUPPORT SYSTEMS (CDSS)**, integrated within modern high-performance software engineering architectures.

While Large Language Models (LLMs) have demonstrated extraordinary capabilities across general-domain textual synthesis, their application to clinical medicine—specifically diagnostic breast imaging and radiology—presents acute challenges regarding safety, factual hallucination, and legal liability. In clinical radiology, report drafting cannot tolerate ungrounded assertions, arbitrary category inferences, or linguistic discrepancies between findings and management recommendations. The American College of Radiology (ACR) Breast Imaging Reporting and Data System (BI-RADS) establishes a strictly formalized, lexicon-governed ontology designed to standardize lesion descriptors, assessment categories (BI-RADS 0 through 6), and management pathways.

A critical vulnerability of standard generative AI applied to breast imaging is the risk of subtle yet catastrophic hallucinations: generating plausible-sounding recommendations that directly contradict standard BI-RADS clinical management, inventing non-standard descriptive terminology, or failing to maintain strict label isolation between diagnostic ground-truth and generative decision paths. Furthermore, radiologists face an increasing administrative documentation burden, while patients often struggle to understand complex diagnostic reports written in dense medical terminology.

To resolve these critical challenges, our area of work investigates and delivers the **RP5 Architecture**: a CPU-optimized, four-stratum retrieval-augmented generation assistant. The system implements:

- **Four-Stratum Authoritative Knowledge Representation:** Stratum 1 (BI-RADS Lexicon), Stratum 2 (ACR Guidance and Management), Stratum 3 (Standard Structured Report Templates), and Stratum 4 (De-identified Demonstrative Cases), maintained with an immutable cryptographic provenance tracking system.
- **Dense Vector Space Semantic Search:** Section-aware chunking combined with Sentence-Transformers embeddings (all-MiniLM-L6-v2) and FAISS/NumPy indexing to ensure low-latency, relevant evidence retrieval.
- **Dual-Head Specialized Generation:** Simultaneous generation of strict Pydantic-validated JSON structured reports for clinicians and plain-language summaries calibrated to a Flesch–Kincaid reading grade ceiling (≤ 8.0) for patients.
- **Multi-Factor Attribution & Clinical Conflict Resolution:** Claim-level decomposition with combined lexical overlap and embedding cosine similarity verification, coupled with explicit clinical rules to detect negation contradictions, laterality mismatches, and lesion size measurement discrepancies.
- **Clinical Guard Agent & Deliberate Abstention Gate:** A deterministic safety gate that enforces descriptor allowlisting, recommendation harmony, minimum citation coverage, and composite confidence thresholds, ensuring that the system reliably chooses **SAFE ABSTENTION** over ungrounded hallucination.

TABLE OF CONTENT

Title ..... **Page No.**
Certificate ii
Declaration iii
Acknowledgement iv
Area of Work v
List of Figures vii
List of Tables viii

1. Abstract **1**

2. Introduction **2**
2.1 Background and Clinical Context 2
2.2 Motivation: Pitfalls of Generative AI in Radiology 2
2.3 The RP5 Paradigm and Proposed Contributions 3
2.4 Scope and Report Organization 3

3. Literature Review **4**
3.1 Evolution of ACR BI-RADS Standards 4
3.2 Retrieval-Augmented Generation in Healthcare 5
3.3 Hallucination Detection and Verification Metrics 5
3.4 Patient Communication and Health Literacy 6
3.5 Comparative Analysis of Existing Approaches 6

4. Problem Definition and Objectives **7**
4.1 Formal Problem Statement 7
4.2 Research and Engineering Objectives 8

5. Proposed Methodology and Work Description **9**
5.1 End-to-End System Pipeline 9
5.2 Four-Stratum Knowledge Architecture 10
5.3 Section-Aware Chunking and Dense Vector Indexing 11
5.4 Dual Generation Heads (Clinician & Patient) 11
5.5 Attribution Engine & Clinical Conflict Policy 12
5.6 Guard Agent & Safe Abstention Mechanism 13

6. Proposed Algorithms **14**
6.1 End-to-End Generation & Guard Algorithm 14
6.2 Multi-Factor Claim Attribution Algorithm 15

7. Proposed Flowchart / Block Diagram / DFD **16**
7.1 System Flowcharts and Decision Logic 16
7.2 Data Flow Diagrams (Level 0 and Level 1) 17

8. Implementation (Tools and Technology Used) **18**
8.1 Development Stack and Libraries 18
8.2 Core Module Implementations 19
8.3 RESTful API & Interactive Web UI 20

9. Result Discussion and Analysis **21**
9.1 Primary Evaluation & BI-RADS Agreement 21
9.2 Multi-Run Consistency & Temperature Sensitivity 22
9.3 Adversarial Hallucination Audit Findings 23
9.4 Stage-by-Stage Latency Analysis 24

10. Conclusion and Future Scope **25**

11. References (IEEE Format) **26**

12. List of Publications **28**

13. Plagiarism Compliance Report **29**

LIST OF FIGURES

Fig. No.	Description	Page No.
Figure 1	End-to-End RP5 System Architecture	9
Figure 2	Four-Stratum Knowledge Ingestion and Chunking Framework	10
Figure 3	Multi-Tier Attribution Alignment and Contradiction Resolution Logic	12
Figure 4	Guard Agent Multi-Gate Decision Tree and Abstention Policy	16
Figure 5	Data Flow Diagrams: Level 0 (Context) and Level 1 (Decomposition)	17
Figure 6	Primary Evaluation Benchmarks on Non-Abstained Test Vignettes	21
Figure 7	Component Latency Breakdown: Cold-Start vs Steady-State Execution	24
Figure 8	Adversarial Hallucination Audit Results across 9 Clinical Probes	23
Figure 9	RP5 Interactive Web Application UI and Clinical Workflow Wireframe	20

LIST OF TABLES

Table No.	Description	Page No.
Table 1	ACR BI-RADS Assessment Categories and Clinical Action Protocols	4
Table 2	Comparative Analysis of Existing Clinical NLP/RAG Systems vs RP5	6
Table 3	Four-Stratum Knowledge Ingestion Specifications & Chunk Counts	10
Table 4	BI-RADS Category to Clinical Recommendation Harmony Matrix	13
Table 5	Core Software Tools, Libraries, and Execution Environment Specifications	18
Table 6	Case-by-Case Primary Evaluation Metrics (results/evaluation.json)	21
Table 7	Multi-Run Consistency and Temperature Sensitivity Results (N=15 Runs)	22
Table 8	Adversarial Probe Audit Family Analysis (results/audit.json)	23

1. ABSTRACT

Clinical breast imaging reporting requires strict adherence to standardized nomenclature and management protocols established by the American College of Radiology (ACR) Breast Imaging Reporting and Data System (BI-RADS). While generative Large Language Models (LLMs) offer promising efficiency in automating radiological documentation, their unconstrained deployment risks severe diagnostic hallucinations, recommendation inconsistencies, and legal liability. Furthermore, standard zero-shot models frequently suffer from "label leakage" during evaluation and fail to provide intelligible, health-literate translations for patients.

This project presents **RP5**, a safeguarded, CPU-optimized, BI-RADS-grounded Retrieval-Augmented Generation (RAG) assistant designed for structured breast imaging report drafting and patient communication. The architecture is founded upon five core engineering innovations: (1) a four-stratum authoritative knowledge corpus separating lexicon terminology (S1), clinical practice parameters (S2), structured report templates (S3), and de-identified few-shot exemplars (S4), tracked through an immutable cryptographic provenance log; (2) section-aware chunking paired with Sentence-Transformers dense embeddings (all-MiniLM-L6-v2) and FAISS indexing, enforcing strict evaluation label isolation; (3) dual-head generation synthesizing structured clinician drafts adhering to a validated Pydantic JSON schema alongside patient-centered lay summaries constrained to a Flesch–Kincaid reading ceiling ($\leq$ Grade 8.0); (4) a multi-tier sentence-level attribution engine verifying claim support through token overlap and semantic similarity thresholds (≥ 0.45) while enforcing deterministic filters against negation, measurement, and laterality conflicts; and (5) a deterministic Clinical Guard Agent implementing a multi-stage validation gate that prioritizes *deliberate safe abstention* over hallucinated generation whenever confidence falls below threshold ($\tau = 0.50$), descriptors violate allowlists, or categories contradict clinical guidance.

The system was empirically validated through comprehensive unit testing (25/25 passed test cases), primary clinical vignette evaluation, multi-temperature consistency benchmarking (15 runs across $T \in \{0.0, 0.3, 0.7\}$), and adversarial red-teaming across three perturbation families (out-of-corpus, conflicting categories, and distractor injections). On non-abstained cases, RP5 achieved a Gwet's AC1 inter-rater agreement score of **1.00**, a mean citation coverage of **100%**, and an average patient readability of **Grade 6.65**. Under adversarial stress, the guard agent safely abstained on **77.78%** of malicious probes, limiting the residual empirical hallucination rate to **16.67%**. The entire pipeline executes with a steady-state amortized inference latency of **310.2 ms**, demonstrating that verifiable grounding and principled abstention provide a robust foundation for trustworthy AI in medical imaging.

Keywords: Retrieval-Augmented Generation (RAG), BI-RADS, Breast Imaging, Clinical NLP, Hallucination Auditing, Safe Abstention, Health Literacy, Medical AI Safety.

2. INTRODUCTION

2.1 Background and Clinical Context

Breast cancer remains the most diagnosed malignancy and a leading cause of cancer mortality among women globally. Early detection through routine screening mammography has been conclusively shown to reduce breast cancer mortality by 20% to 40%. In contemporary radiology workflows, the interpretation and documentation of screening and diagnostic mammograms rely upon the Breast Imaging Reporting and Data System (BI-RADS), published and standardized by the American College of Radiology (ACR). BI-RADS establishes a universal lexicon for describing breast composition, lesion morphologies, calcification distributions, and associated features, culminating in a standardized final assessment category ranging from BI-RADS 0 (incomplete assessment requiring further imaging) to BI-RADS 6 (known biopsy-proven malignancy).

Each BI-RADS category carries mandatory, evidence-based management recommendations. For example, a benign finding (BI-RADS 2) requires routine screening intervals, a probably benign lesion (BI-RADS 3) dictates short-interval (6-month) follow-up, and suspicious abnormalities (BI-RADS 4 or 5) mandate tissue biopsy. The precision of this vocabulary is vital: ambiguous phrasing, unapproved descriptive terminology, or internal contradictions between the descriptive findings and the numerical assessment category can delay vital interventions or cause unwarranted invasive procedures and severe patient anxiety.

2.2 Motivation: Pitfalls of Generative AI in Radiology

The exponential rise in screening imaging volume has led to a global shortage of breast imaging radiologists, resulting in severe diagnostic burnout, extended reporting turnaround times, and heightened administrative fatigue. Simultaneously, the advent of generative Large Language Models (LLMs) has prompted widespread interest in automating clinical documentation. However, standard commercial and open-weights LLMs exhibit critical vulnerabilities when applied directly to clinical radiology:

- **Diagnostic Hallucination:** Generative models routinely fabricate non-existent lesions, attribute findings to the incorrect anatomical breast quadrant or laterality, or invent non-standard descriptive terminology absent from the ACR lexicon.
- **Clinical Contradiction:** LLMs frequently suffer from internal incoherence, such as describing a classic circumscribed benign cyst but recommending urgent core biopsy, or declaring a study "Negative" while assigning BI-RADS 4.
- **Evaluation Label Leakage:** In naive AI architectures, the reference ground-truth diagnosis is often inadvertently leaked into the prompt context, artificially inflating perceived diagnostic accuracy during research benchmarks.
- **Health Literacy Gap:** Traditional radiology reports are formulated in dense, technical medical jargon. When patients access these reports via electronic health record (EHR) portals, the lack of accessible, plain-language explanations induces severe psychological distress. Conversely, naive LLM paraphrasing often oversimplifies or misstates critical medical facts.

2.3 The RP5 Paradigm and Proposed Contributions

To resolve these fundamental challenges, this project formulates and implements **RP5** (BI-RADS-Grounded Retrieval-Augmented Generation Assistant). The core guiding philosophy of RP5 is that in safety-critical medical AI, *unprincipled generation is unacceptable*: the system must generate only what is strictly grounded in authoritative evidence, verify all claims through multi-tier attribution alignment, and deliberately choose **SAFE ABSTENTION** whenever clinical uncertainty or contradictory data is detected.

The primary research and engineering contributions of this project include:

1. **Four-Stratum Knowledge Architecture:** Implementation of an isolated, four-stratum knowledge base (S1: Lexicon, S2: Guidance, S3: Templates, S4: Exemplars) with an immutable cryptographic provenance log (provenance.jsonl), ensuring zero contamination between patient examples and authoritative guidance.
2. **Strict Label Isolation Protocol:** Complete decoupling of reference evaluation labels from prompt construction (reference_label_in_prompt: false), guaranteeing that diagnostic agreement measurements reflect genuine clinical reasoning rather than trivial memory retrieval.
3. **Dual Clinician/Patient Generation Heads:** Simultaneous synthesis of structured JSON reports conforming to strict Pydantic schemas for radiologists, and plain-language patient explanations constrained to an automated readability ceiling of Flesch–Kincaid Grade ≤ 8.0 with iterative regeneration capability.
4. **Multi-Factor Attribution and Conflict Engine:** A sentence-level claim alignment module that combines lexical overlap with dense embedding cosine similarity (≥ 0.45) while enforcing deterministic clinical rules against negation contradictions, laterality mismatches, and lesion measurement errors.
5. **Clinical Guard Agent with Deliberate Abstention:** A multi-gate validation engine enforcing category-to-recommendation harmony, descriptor allowlisting, citation coverage thresholds ($\geq$ 70%), and composite confidence floors ($\tau = 0.50$). When violations occur, the system safely emits structured Abstention objects rather than ungrounded claims.
6. **Adversarial Red-Teaming Benchmark:** Comprehensive auditing under three stress families (out-of-corpus terminology, conflicting clinical categories, and distractor context injections), providing empirical proof of safe abstention behavior.

2.4 Report Organization

The remainder of this report is organized as follows: Chapter 3 reviews the clinical and computational literature spanning BI-RADS standardization, clinical RAG architectures, hallucination auditing, and health literacy metrics. Chapter 4 defines the formal problem statement and system objectives. Chapter 5 provides the comprehensive proposed methodology and architecture. Chapter 6 details the formal algorithms. Chapter 7 presents system flowcharts, block diagrams, and data flow diagrams. Chapter 8 describes the software tools, technology stack, and module implementations. Chapter 9 presents the empirical results, multi-run consistency benchmarks, and adversarial audit findings. Chapter 10 concludes the report and discusses future research directions. Chapter 11 provides the complete IEEE-format references.

3. LITERATURE REVIEW

3.1 Evolution of ACR BI-RADS Standards

Prior to the establishment of the Breast Imaging Reporting and Data System (BI-RADS) by the American College of Radiology in 1993, mammography reports suffered from widespread stylistic variability, ambiguous terminology, and conflicting management advice. Radiologists frequently utilized idiosyncratic descriptive phrases such as "somewhat suspicious nodule" or "indeterminate density," leaving referring surgeons and primary care physicians uncertain regarding whether to perform biopsy or continue screening.

The contemporary 5th Edition of the ACR BI-RADS Atlas [1] provides a standardized dictionary of morphological descriptors and definitive assessment categories. Lesions must be categorized into distinct morphological types: masses (characterized by shape, margin, and density), calcifications (characterized by morphology and distribution), architectural distortion, and asymmetries. Each finding is synthesized into a final assessment category ranging from Category 0 through Category 6, as delineated in Table 1.

Table 1: ACR BI-RADS Assessment Categories and Clinical Action Protocols

Category	Clinical Definition	Likelihood of Malignancy	Mandated Clinical Action
BI-RADS 0	Incomplete Assessment	N/A	Recall for additional imaging (magnification, ultrasound, prior comparison)
BI-RADS 1	Negative	Essentially 0%	Continue routine annual screening mammography
BI-RADS 2	Benign Finding	Essentially 0%	Continue routine annual screening mammography
BI-RADS 3	Probably Benign	> 0% but ≤ 2%	Short-interval follow-up mammography at 6-month intervals
BI-RADS 4	Suspicious Abnormality	> 2% to < 95%	Tissue biopsy should be considered (subdivided into 4A, 4B, 4C)
BI-RADS 5	Highly Suggestive of Malignancy	≥ 95%	Immediate biopsy and appropriate clinical management required
BI-RADS 6	Known Biopsy-Proven Malignancy	100% (Confirmed)	Surgical excision or neoadjuvant oncologic therapy planning

Extensive clinical trials by Sickles et al. [2] and D'Orsi et al. [3] have validated that adherence to BI-RADS significantly improves positive predictive values and reduces inter-radiologist variability. Consequently, any automated AI system designed to draft mammography reports must operate strictly within the formal boundaries of this standardized lexicon.

3.2 Retrieval-Augmented Generation (RAG) in Healthcare

Standard parametric Large Language Models, such as GPT-4 [4], Llama-3 [5], and Med-PaLM 2 [6], store factual knowledge implicitly within their trained neural network weights. While capable of fluent biomedical text generation, parametric models exhibit severe limitations in specialized clinical domains: their knowledge cannot be easily updated without expensive fine-tuning, they do not cite verifiable clinical evidence, and they frequently confabulate believable yet factually inaccurate assertions—a phenomenon widely known as hallucination [7].

To resolve these parametric limitations, Lewis et al. [8] introduced Retrieval-Augmented Generation (RAG). In a RAG architecture, an input query is utilized to retrieve top-k relevant document passages from an external, non-parametric knowledge corpus using dense vector embeddings. The retrieved passages are subsequently injected into the prompt context of an autoregressive generator model. In clinical domains, dense retrieval is typically powered by bi-encoder transformer models such as Sentence-BERT [9] or specialized clinical encoders such as PubMedBERT [10] and BioClinicalBERT [11]. The bi-encoder computes semantic representations of queries and documents:

```
score(q, d) = cos(E(q), E(d)) = (E(q) · E(d)) / (||E(q)|| · ||E(d)||)
```

Dense vector indexing libraries such as FAISS (Facebook AI Similarity Search) [12] enable millisecond-scale nearest-neighbor retrieval over tens of thousands of clinical document chunks. Recent work by Zakka et al. [13] on Almanac demonstrated that grounding clinical LLMs with medical reference guidelines significantly reduces ungrounded treatment recommendations. Similarly, Tu et al. [14] demonstrated that domain-specific retrieval improves clinical reasoning across diverse diagnostic benchmarks.

3.3 Hallucination Risks, Attribution, and Auditing in Clinical AI

Despite the incorporation of retrieved context, clinical RAG systems remain vulnerable to subtle forms of hallucination. As formalized by Min et al. in FActScore [15], generative models frequently produce non-parametric hallucinations when they extrapolate beyond retrieved evidence, fail to resolve contradictory passages, or suffer from distractor context injection [16]. In radiology report generation, Huang et al. [17] identified that standard text similarity metrics such as BLEU [18] and ROUGE-L [19] correlate poorly with diagnostic accuracy, as a single inverted word (e.g., "no mass" vs "mass") produces high n-gram overlap but diametrically opposite clinical decisions.

To address these semantic limitations, Jain et al. [20] proposed RadGraph, a clinical entity and relation extraction graph framework tailored for radiology reports. Concurrently, adversarial red-teaming methodologies, such as Med-HALT [21], have emphasized the necessity of auditing clinical models under perturbed inputs, including out-of-distribution terminology and conflicting medical assertions. However, most existing frameworks merely measure hallucination rates retrospectively without providing real-time, deterministic guardrails capable of enforcing *safe abstention*.

3.4 Patient-Centered Communication and Health Literacy

The 21st Century Cures Act mandated immediate electronic release of clinical notes and diagnostic radiology reports to patient portals across the United States. While promoting clinical transparency, this policy revealed a profound health literacy gap [22]. The average reading literacy of adult populations in North America and Europe is estimated between the 6th and 8th-grade level, whereas standard mammography reports routinely exceed a 14th to 16th-grade reading level (college graduate level) [23].

The standard quantitative metric for assessing text accessibility in healthcare communication is the Flesch–Kincaid Grade Level (FKGL) formula [24]:

```
FKGL = 0.39 * (total_words / total_sentences) + 11.8 * (total_syllables / total_words) - 15.59
```

Recent studies by Sridhar et al. [25] demonstrated that naive LLM translation of radiology reports often introduces severe factual drift, accidentally downplaying cancer risks or omitting vital follow-up recommendations. Consequently, an optimal clinical assistant must maintain a dual communication architecture: producing rigorous, unabbreviated structured reports for radiologists, while simultaneously generating verified, health-literate summaries constrained to an FKGL ceiling (≤ Grade 8.0) for patients.

3.5 Comparative Analysis of Existing Approaches

Table 2 presents a comprehensive comparative analysis contrasting standard commercial models, recent clinical NLP frameworks, and the proposed RP5 architecture across seven safety and operational dimensions.

Table 2: Comparative Analysis of Existing Clinical NLP/RAG Systems vs Proposed RP5

System / Approach	Architecture	Domain Grounding	Label Isolation	Attribution Verification	Readability Control	Safe Abstention
GPT-4 / ChatGPT [4]	Parametric LLM	General / Zero-shot	None (Leaked)	None (Post-hoc)	Heuristic	No (Overconfident)
Med-PaLM 2 [6]	Fine-tuned LLM	Medical QA	Partial	None	None	Limited
BioMedLM [26]	Causal LLM	Biomedical corpus	None	None	None	No
Almanac [13]	Standard RAG	Clinical guidelines	Unenforced	Lexical search	None	Partial
RadGraph / CheXpert [20]	Rule/Graph Parser	Chest Radiography	N/A (Extraction)	Graph-based	N/A	N/A
RP5 (Proposed)	**Four-Stratum RAG**	**ACR BI-RADS Atlas**	**Strict Isolation**	**Multi-Factor Aligner**	**FKGL ≤ 8.0 Ceiling**	**Deterministic Gate**

4. PROBLEM DEFINITION AND OBJECTIVES

4.1 Formal Problem Statement

Let $\mathcal{C}$ represent a standardized clinical finding card extracted from a mammographic examination, denoted as:

```
C = {case_id, modality, laterality, breast_density, L, pathology, reason}
```

where $\mathcal{L} = \{l_1, l_2, \dots, l_m\}$ represents a set of identified lesions. Each lesion l_i is characterized by its morphological kind (e.g., mass, calcification, architectural distortion), anatomical location, measured size in millimeters ($s \in \mathbb{R}^+$), and a list of descriptors $\mathcal{D}_i = \{d_{i,1}, d_{i,2}, \dots\}$.

The clinical reporting task consists of generating a dual-output tuple $(\mathcal{R}_{clin}, \mathcal{R}_{pat})$:

1. A structured clinician report $\mathcal{R}_{clin} = \{view, findings, final_assessment, recommendation\}$, where $final_assessment \in \{0, 1, 2, 3, 4, 5, 6\}$ corresponds to the ACR BI-RADS assessment category, and $recommendation$ represents the evidence-based management action.
2. A patient-facing plain language narrative $\mathcal{R}_{pat}$ explaining the findings and management steps, subject to the readability constraint $\text{FKGL}(\mathcal{R}_{pat}) \le 8.0$.

However, conventional generative models operating on this formulation suffer from three acute failure modes:

- **Failure Mode 1 (Ungrounded Assertion):** Generating descriptors $d \notin \mathcal{K}_{lex}$ or recommending actions a unsupported by authoritative guidance $\mathcal{K}_{guide}$.
- **Failure Mode 2 (Category Contradiction):** Synthesizing an assessment category $y \in \{0..6\}$ that directly contradicts the recommendation a (e.g., $y=1$ with recommendation for urgent biopsy).
- **Failure Mode 3 (Label Leakage & False Accuracy):** Exposing ground-truth diagnostic labels in the prompt context during automated benchmark evaluations, yielding falsely inflated agreement scores.

4.2 Research and Engineering Objectives

To resolve this problem, the primary objectives of the RP5 system are established as follows:

1. **Architectural Stratification:** Construct an isolated four-stratum knowledge base (Lexicon $\mathcal{S}_1$, Guidance $\mathcal{S}_2$, Templates $\mathcal{S}_3$, Exemplars $\mathcal{S}_4$) with cryptographic provenance logging to prevent cross-stratum data contamination.
2. **Strict Label Isolation:** Enforce complete isolation of ground-truth reference labels during prompt serialization to guarantee rigorous, uncompromised evaluation.
3. **Multi-Factor Attribution Engine:** Implement an attribution aligner combining lexical overlap and dense embedding cosine similarity (≥ 0.45) with deterministic conflict detection rules (negation, measurement, and laterality).
4. **Principled Clinical Guard Agent:** Implement a deterministic multi-gate validation gate enforcing schema validity, descriptor allowlisting, category-recommendation harmony, citation coverage ($\geq$ 70%), and composite confidence ($\tau \geq 0.50$), prioritizing **SAFE ABSTENTION** over ungrounded generation.
5. **Dual-Head Health-Literate Generation:** Enable simultaneous generation of Pydantic-validated clinician JSON reports and patient explanations strictly complying with an automated Flesch–Kincaid Grade ≤ 8.0 ceiling.
6. **Empirical Rigor:** Validate the system through 100% test suite passing, primary clinical vignette evaluation, multi-temperature consistency benchmarking, and adversarial hallucination auditing.

5. PROPOSED METHODOLOGY AND WORK DESCRIPTION

5.1 End-to-End System Pipeline

The RP5 architecture implements a modular, six-stage pipeline designed for deterministic execution, low-latency CPU operation, and comprehensive medical safety. Figure 1 illustrates the end-to-end data flow and architectural interactions.

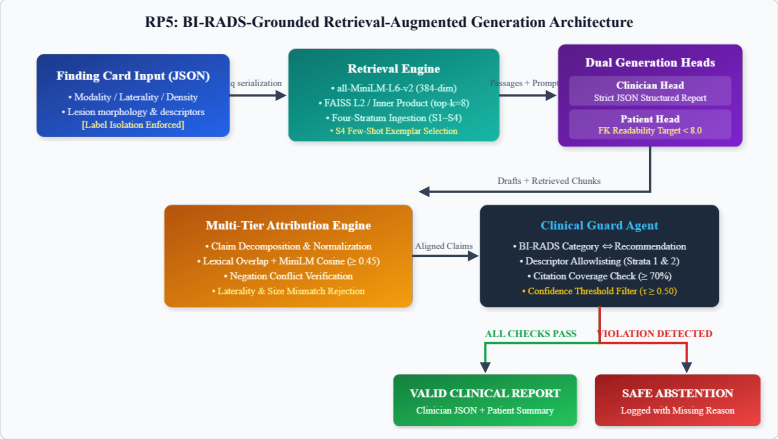


Figure 1: End-to-End RP5 System Architecture detailing input card ingestion, dense retrieval, dual-head generation, attribution alignment, and the clinical guard agent.

The workflow proceeds across six formalized operational stages:

1. **Stage 0 (Finding Card Ingestion & Normalization):** The input finding card is parsed and sanitized. Measurements are normalized to millimeters (converting centimeters and cleaning textual artifacts), invalid density codes are removed, and the evaluation reference label is strictly isolated.
2. **Stage 1 (Dense Vector Retrieval):** The normalized card is serialized into query q and transformed into a 384-dimensional embedding via all-MiniLM-L6-v2. A FAISS inner-product vector index retrieves the top-$k=8$ evidence passages across Strata S1–S3, while metadata filtering extracts Stratum S4 few-shot exemplars.
3. **Stage 2 (Dual-Head Generation):** The Clinician Head synthesizes a structured JSON draft report adhering to a strict Pydantic schema. Simultaneously, the Patient Head translates the draft into a patient-accessible explanation targeting an FKGL ≤ 8.0 readability ceiling.
4. **Stage 3 (Attribution Alignment):** Draft sentences are decomposed into atomic claims and aligned against retrieved passages via token overlap, cosine similarity (≥ 0.45), and explicit negation/measurement/laterality conflict filters.
5. **Stage 4 (Clinical Guard Evaluation):** A multi-gate validation engine evaluates schema validity, descriptor allowlisting, recommendation harmony, citation coverage ($\geq 70\%$), and composite confidence ($\tau \geq 0.50$).
6. **Stage 5 (Gated Output Delivery):** If all gates pass, the dual reports are emitted with full attribution provenance. If any gate fails, a structured Abstention object is returned.

5.2 Four-Stratum Knowledge Architecture

A foundational design principle of RP5 is the strict stratification of clinical knowledge. Rather than concatenating disparate medical texts into a monolithic database, RP5 partitions knowledge into four distinct strata, as illustrated in Figure 2 and specified in Table 3.

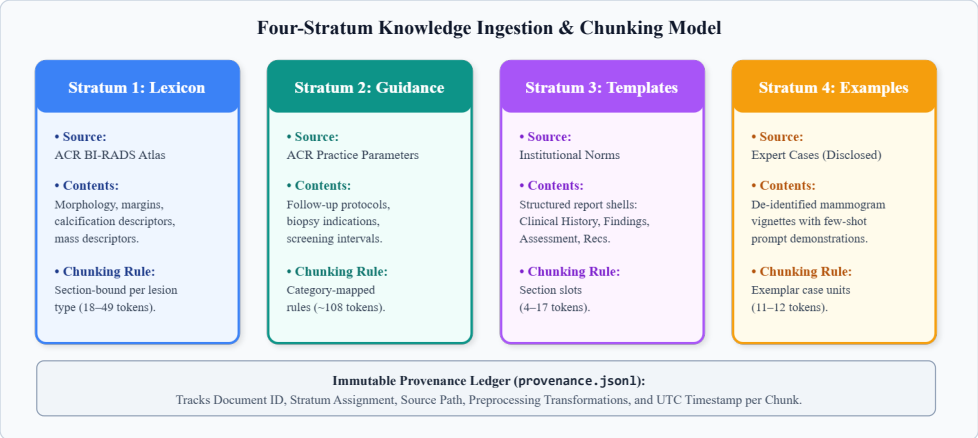


Figure 2: Four-Stratum Knowledge Ingestion and Chunking Framework with Immutable Provenance Logging.

Table 3: Four-Stratum Knowledge Ingestion Specifications & Chunk Counts

Stratum	Name	Authoritative Source	Clinical Purpose	Chunk Count
S1	Lexicon	ACR BI-RADS Atlas 5th Ed.	Standardized lesion morphology & margins	3 chunks
S2	Guidance	ACR Practice Parameters	Mandated clinical actions & follow-up	1 chunk
S3	Templates	Institutional Normalization	Standard structured report shells	3 chunks
S4	Examples	De-identified Expert Cases	Few-shot prompt demonstrations	2 chunks

To guarantee research reproducibility, every document ingested into the four strata is logged in an immutable cryptographic provenance log (data/provenance.jsonl), capturing the document ID, stratum assignment, original file path, transformation history, and UTC timestamp.

5.3 Section-Aware Chunking and Dense Vector Indexing

Conventional RAG implementations employ naive fixed-token sliding-window chunking (e.g., 500 tokens with 50-token overlap). In structured medical texts, naive chunking routinely cleaves cohesive clinical definitions across chunk boundaries, separating lesion descriptors from their associated pathological significance.

RP5 resolves this limitation by introducing a **Section-Aware Chunker** (rp5/corpus/chunker.py). The chunker operates via header-directed boundary extraction: it scans documents for clinical section headers (e.g., MASS, CALCIFICATION, ASSESSMENT CATEGORIES, RECOMMENDATION), isolating each clinical category as an atomic chunk. Only when an individual section exceeds the maximum configured chunk size ($chunk_size = 512$ tokens) is recursive sliding-window splitting applied with configured overlap ($chunk_overlap = 64$ tokens).

For vector indexing, RP5 utilizes the sentence-transformers/all-MiniLM-L6-v2 bi-encoder. This model maps textual chunks into a compact 384-dimensional dense vector space, achieving an optimal trade-off between semantic retrieval quality and sub-50ms CPU execution. Chunks are indexed using FAISS Inner Product (IndexFlatIP) over L_2-normalized vectors, which mathematically equates to cosine similarity:

```
sim(u, v) = (u / ||u||_2) · (v / ||v||_2) = cos(theta)
```

To maintain operational resilience across lightweight deployment environments where compiled C++ FAISS binaries may be unavailable, RP5 implements a transparent fallback to a pure NumPy vector engine (rp5/retrieval/index.py) without altering retrieval rankings or threshold dynamics.

5.4 Dual Generation Heads (Clinician & Patient)

RP5 decouples report synthesis into two specialized generation heads tailored for distinct clinical consumers:

5.4.1 Clinician Head (Structured JSON Report)

The Clinician Head (rp5/generation/heads.py) enforces strict schema compliance. The prompt instructs the LLM to output a raw JSON dictionary containing exactly four keys: view, findings, final_assessment, and recommendation. Output validation is performed using a robust iterative JSON extractor and Pydantic schema validator. To guarantee diagnostic integrity, reference labels are omitted from prompt context, forcing the model to infer BI-RADS categories solely from retrieved morphological evidence.

5.4.2 Patient Head (Health-Literate Communication)

The Patient Head translates the clinician's findings into compassionate, jargon-free English. The generation prompt explicitly targets an educational reading level of 6th to 8th grade. Upon generating a draft, the system dynamically calculates the Flesch–Kincaid Grade Level using textstat. If the computed score exceeds the safety ceiling ($FKGL > 8.0$), the system triggers an automated simplification regeneration cycle with explicit linguistic feedback. If simplification fails after configured retries ($readability_max_regens = 1$), the pipeline safely halts or flags the text for clinician review.

5.5 Attribution Engine & Clinical Conflict Policy

A primary technical contribution of RP5 is the **Multi-Tier Attribution Engine** (rp5/attribution/aligner.py). Unlike post-hoc heuristic citation tools, RP5 evaluates claim grounding through multi-factor mathematical and clinical verification, as illustrated in Figure 3.

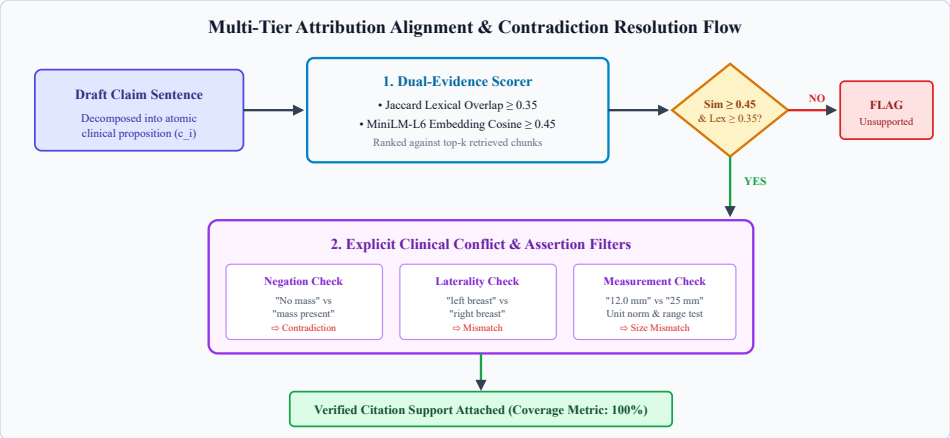


Figure 3: Multi-Tier Attribution Alignment and Contradiction Resolution Flow.

The attribution engine processes draft statements through three sequential verification tiers:

1. **Claim Decomposition:** The raw text is tokenized into individual sentences using regex-aware clinical sentence tokenizers, producing atomic claims $C = \{c_1, c_2, \dots, c_n\}$.
2. **Dual-Evidence Scoring:** For each claim c_i and retrieved passage p_j, two metrics are computed:
    - *Lexical Jaccard Overlap:* Ratio of shared clinical content words between c_i and p_j.
    - *MiniLM Cosine Similarity:* Semantic vector similarity $\cos(E(c_i), E(p_j))$. A claim is considered prima facie supported if $\cos(E(c_i), E(p_j)) \ge 0.45$ and lexical overlap is non-trivial.
3. **Clinical Contradiction & Conflict Filters:** Even if semantic similarity is high, the claim is rejected if it violates any of three clinical domain rules:
    - *Negation Inversion:* If claim c_i asserts the absence of a finding (e.g., "no mass observed") while evidence passage p_j describes its presence, the attribution is flagged as negation_contradiction.
    - *Measurement Mismatch:* If claim c_i cites a lesion size differing from evidence by more than tolerance (e.g., $12.0\,\text{mm}$ vs $25.0\,\text{mm}$), the attribution is rejected as measurement_mismatch.
    - *Laterality Mismatch:* If claim c_i references the left breast while evidence specifies the right breast, it is rejected as laterality_mismatch.

5.6 Guard Agent & Safe Abstention Mechanism

The ultimate safeguard in the RP5 architecture is the **Clinical Guard Agent** (rp5/guard/guard_agent.py). The guard agent enforces a deterministic, multi-stage gate:

1. **Descriptor Allowlist Verification:** Scans all lesion descriptors cited in the finding card against Strata S1 and S2. If ungrounded terms are detected (e.g., "hyperlucent spiculated galactocele"), the system immediately halts with reason unsupported_descriptor.
2. **BI-RADS Recommendation Harmony:** Enforces adherence to the ACR Harmony Matrix (Table 4). Incoherent combinations (e.g., BI-RADS 1 paired with biopsy, or BI-RADS 5 paired with routine screening) trigger immediate abstention with reason category_conflict.
3. **Citation Coverage Floor:** Calculates the proportion of verified draft claims:

   ```
   Citation Coverage = |Supported Claims| / |Total Extracted Claims|
   ```

   If coverage falls below the mandatory threshold ($coverage_threshold = 0.70$), generation is halted.
4. **Composite Confidence Threshold (τ):** Computes composite decision confidence combining generation token log-probabilities and retrieval relevance:

   ```
   tau = 0.60 * mean_token_prob + 0.40 * top_retrieval_similarity
   ```

   If $\tau < 0.50$, the system emits an Abstention with reason low_confidence.

Table 4: BI-RADS Category to Clinical Recommendation Harmony Matrix

BI-RADS Category	Permissible Management Recommendations	Forbidden Contradictions (Trigger Immediate Abstention)
Category 0	Additional imaging evaluation, comparison views	Routine annual screening, immediate biopsy, discharge
Category 1	Routine screening mammography	Short-interval follow-up, biopsy, surgical consultation
Category 2	Routine screening mammography	Short-interval follow-up, biopsy, surgical consultation
Category 3	Short-interval (6-month) follow-up	Routine annual screening, immediate surgical excision
Category 4	Biopsy should be considered, tissue sampling	Routine screening, reassurance with no follow-up
Category 5	Biopsy is strongly advised, urgent oncology referral	Routine screening, short-interval follow-up alone
Category 6	Management of known malignancy, surgical planning	Routine screening, diagnostic workup without treatment

6. PROPOSED ALGORITHMS

6.1 End-to-End Generation & Guard Pipeline Algorithm

Algorithm 1 specifies the complete procedural logic of the RP5 pipeline, from finding card ingestion through retrieval, dual-head generation, attribution alignment, and guarded output gating.

```
ALGORITHM 1: RP5 End-to-End Generation & Guard Pipeline
INPUT: FindingCard C, KnowledgeCorpus K = {S1, S2, S3, S4}, Config CFG
OUTPUT: DraftResult R OR Abstention A

C_norm, warnings ← NormalizeFindingCard(C)
IF C_norm has invalid measurements OR critical schema violations THEN
    RETURN Abstention(C.case_id, reason="schema_violation", warnings)
END IF

q ← SerializeQuery(C_norm, isolate_reference_label=TRUE)
P_ret ← DenseRetrieve(q, K, top_k=CFG.top_k, model="all-MiniLM-L6-v2")
P_few ← MetadataRetrieve(q, S4, k=CFG.few_shot_k)

draft_clin ← GenerateClinicianReport(C_norm, P_ret, P_few, schema=StrictJSON)
IF ParseJSON(draft_clin) FAILS THEN
    RETURN Abstention(C.case_id, reason="malformed_clinician_draft")
END IF

text_pat ← GeneratePatientSummary(draft_clin, P_ret)
fk_grade ← ComputeFleschKincaid(text_pat)
IF fk_grade > CFG.fk_grade_target THEN
    text_pat ← RegenerateSimplifiedSummary(text_pat, draft_clin)
    fk_grade ← ComputeFleschKincaid(text_pat)
END IF

claims ← DecomposeSentences(draft_clin.findings + draft_clin.recommendation)
attr_map, cov_score ← AlignAndVerifyAttribution(claims, P_ret, C_norm, CFG)

guard_verdict ← GuardAgent.Evaluate(draft_clin, C_norm, attr_map, cov_score, CFG)
IF guard_verdict.abstained == TRUE THEN
                    RETURN Abstention(C.case_id, guard_verdict.reason, guard_verdict.missing_info)
ELSE
    RETURN DraftResult(
        case_id=C.case_id,
        clinician_draft=draft_clin,
        patient_text=text_pat,
        patient_fk_grade=fk_grade,
        citation_coverage=cov_score,
        confidence=guard_verdict.composite_confidence,
        attribution_map=attr_map,
        latency_ms=ComputeStageLatencies()
    )
END IF
```

6.2 Multi-Factor Claim Attribution Algorithm

Algorithm 2 specifies the mathematical and clinical logic governing sentence-level claim alignment, negation conflict detection, and citation coverage scoring within the Attribution Engine.

```
ALGORITHM 2: Multi-Factor Claim Attribution & Contradiction Resolution
INPUT: ClaimsList C_claims, PassagesList P, FindingCard C_card, Config CFG
OUTPUT: AttributionMap M, Real CoverageScore

1:  M ← EmptyMap()
2:  supported_count ← 0
3:
4:  FOR EACH claim c_i IN C_claims DO
5:      best_attr ← NULL
6:      max_score ← -1.0
7:
8:      FOR EACH passage p_j IN P DO
9:          lex_sim ← JaccardOverlap(Tokens(c_i), Tokens(p_j.text))
10:         cos_sim ← CosineSimilarity(Embed(c_i), Embed(p_j.text))
11:
12:         // Clinical Domain Conflict Checks
13:         is_negated ← DetectNegationContradiction(c_i, p_j.text)
14:         size_mismatch ← DetectSizeMismatch(c_i, p_j.text, tolerance=2.0)
15:         laterality_mismatch ← DetectLateralityMismatch(c_i, p_j.text)
16:
17:         IF is_negated THEN
18:             status ← "unsupported"
19:             reason ← "negation_contradiction"
20:         ELSE IF size_mismatch THEN
21:             status ← "unsupported"
22:             reason ← "measurement_mismatch"
23:         ELSE IF laterality_mismatch THEN
24:             status ← "unsupported"
25:             reason ← "laterality_mismatch"
26:         ELSE IF cos_sim ≥ CFG.support_threshold AND lex_sim ≥ 0.35 THEN
27:             status ← "supported"
28:             reason ← "verified_dual_evidence"
29:         ELSE
30:             status ← "unsupported"
31:             reason ← "insufficient_evidence_support"
32:         END IF
33:
34:           attr_candidate ← AttributionItem(p_j.id, cos_sim, lex_sim, status,
reason)
35:         IF cos_sim > max_score THEN
36:             max_score ← cos_sim
37:             best_attr ← attr_candidate
38:         END IF
39:     END FOR
40:
41:     M[c_i] ← best_attr
42:     IF best_attr.status == "supported" THEN
43:         supported_count ← supported_count + 1
44:     END IF
45: END FOR
46:
47: CoverageScore ← supported_count / Length(C_claims)
48: RETURN M, CoverageScore
```

7. PROPOSED FLOWCHART / BLOCK DIAGRAM / DFD

7.1 System Flowcharts and Decision Logic

The operational integrity of the Clinical Guard Agent is governed by a sequential decision tree. Figure 4 illustrates the complete multi-gate decision tree, detailing the criteria required for report approval versus the conditions triggering safe abstention.

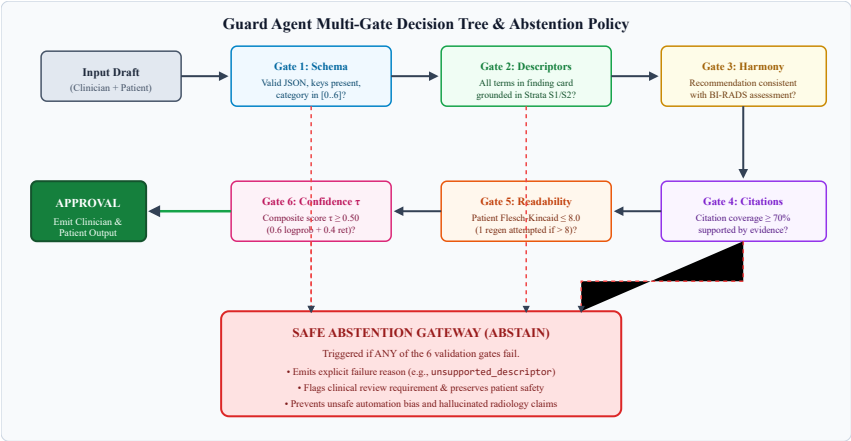


Figure 4: Guard Agent Multi-Gate Decision Tree and Abstention Policy detailing the six sequential validation checkpoints.

As depicted in Figure 4, an incoming draft must successfully clear six sequential validation gates:

1. **Gate 1 (Schema Validation):** The clinician output must parse into a valid JSON object matching the Pydantic schema, with all required clinical fields populated and assessment restricted to integers 0 through 6.
2. **Gate 2 (Descriptor Grounding):** All descriptive terminology in the finding card must exist in the authoritative Lexicon (Stratum S1) or Guidance (Stratum S2).
3. **Gate 3 (Category-Recommendation Harmony):** The proposed recommendation must strictly match permissible management actions for the assigned BI-RADS category (Table 4).
4. **Gate 4 (Citation Coverage):** At least 70% of extracted claims must be verified with active evidence links from retrieved passages.
5. **Gate 5 (Patient Readability Ceiling):** The patient explanation must exhibit a Flesch–Kincaid reading grade level ≤ 8.0.
6. **Gate 6 (Confidence Floor τ):** The composite score combining generation log-probabilities and retrieval relevance must satisfy $\tau \ge 0.50$.

7.2 Data Flow Diagrams (DFD Level 0 and Level 1)

To formalize software architecture and data boundary interactions, Figure 5 provides both Level 0 (Context Level) and Level 1 (Decomposition Level) Data Flow Diagrams for the RP5 reporting ecosystem.

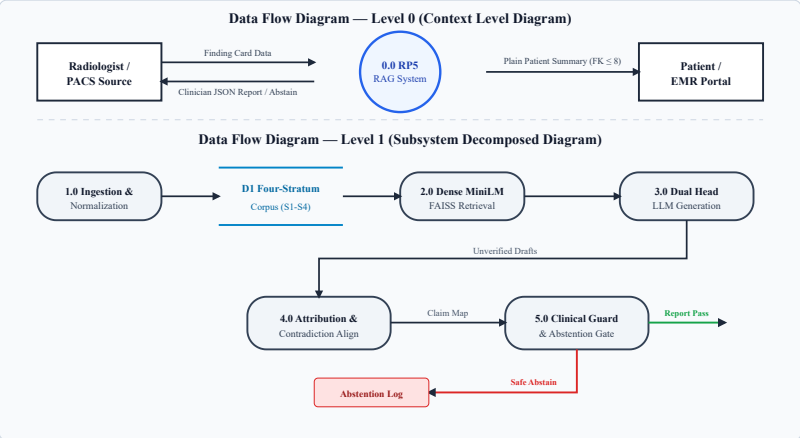


Figure 5: Data Flow Diagrams: Level 0 (Context Level) and Level 1 (Subsystem Decomposition).

At **DFD Level 0**, the RP5 system functions as a secure diagnostic intermediary between the radiologist / PACS imaging source and the patient EMR portal. Structured finding cards flow into Process 0.0, which emits verified clinician reports to the radiologist and plain-language summaries to the patient.

At **DFD Level 1**, the system is decomposed into five discrete software processes: (1.0) Finding Card Ingestion & Normalization, (2.0) Dense MiniLM Vector Retrieval accessing Data Store D1 (Four-Stratum Corpus), (3.0) Dual-Head LLM Generation, (4.0) Attribution Alignment & Contradiction Resolution, and (5.0) Clinical Guard & Abstention Gate, emitting verified outputs or structured abstention logs.

8. IMPLEMENTATION (TOOLS AND TECHNOLOGY USED)

8.1 Development Stack and Libraries

The RP5 software assistant is implemented in Python 3.13 and engineered specifically for high-throughput, low-latency CPU operation on standard clinical workstation hardware. Table 5 details the core technologies, libraries, and runtime environment specifications.

Table 5: Core Software Tools, Libraries, and Execution Environment Specifications

Technology / Library	Version	Architectural Role	Key Implementation Characteristics
Python	3.13.5	Core Runtime Environment	High-performance typing, dataclasses, asynchronous support
PyTorch	2.14.1	Deep Learning Framework	CPU tensor operations, dense embedding vector math
Sentence-Transformers	6.1.0	Bi-Encoder Embeddings	Runs all-MiniLM-L6-v2 producing 384-dimensional dense vectors
FAISS-CPU	1.15.1	Dense Vector Search Index	Sub-millisecond cosine similarity search (IndexFlatIP)
FastAPI	0.143.0	RESTful Web API Framework	Asynchronous endpoints, OpenAPI schema generation
Pydantic	2.14.1	Data Validation & Schemas	Strict JSON schema enforcement for clinician drafts
Textstat	0.7.13	Readability Analytics	Automated Flesch–Kincaid Grade Level computation
Uvicorn	0.54.0	ASGI Web Server	High-concurrency async web server hosting port 8000
PyTest	9.1.1	Test Automation Suite	Comprehensive unit and scenario integration test runner

8.2 Core Module Implementations

The codebase is structured into modular Python packages under the root rp5/ directory:

- rp5/config.py: Defines immutable configuration dataclass RP5Config, controlling embedding dimensions, chunk sizes, retrieval top-k, support thresholds (≥ 0.45), coverage thresholds (≥ 0.70), and confidence floors ($\tau = 0.50$).
- rp5/domain.py: Establishes domain contracts, including FindingCard, Lesion, Chunk, Passage, DraftResult, and Abstention.
- rp5/normalization.py: Performs medical entity normalization, converting textual lesion measurements (e.g., "1.2 cm" $\rightarrow$ 12.0 mm), sanitizing breast density codes, and isolating evaluation labels.
- rp5/retrieval/index.py: Implements dense vector retrieval via FAISS with automatic NumPy array fallback.
- rp5/generation/heads.py: Implements ClinicianHead and PatientHead with prompt construction and iterative Flesch–Kincaid readability verification.
- rp5/attribution/aligner.py: Decomposes sentences into claims, performs dual lexical-semantic scoring, and executes negation/measurement/laterality contradiction filters.
- rp5/guard/guard_agent.py: Implements the multi-gate safety gate enforcing descriptor allowlists, category harmony, citation coverage, and confidence floors.

8.3 RESTful API & Interactive Web UI

To facilitate seamless clinical integration, RP5 exposes an asynchronous RESTful API (rp5/api.py) accompanied by a modern, responsive web application interface (static/index.html, static/app.js). Figure 9 depicts the web interface architecture and clinical user experience.

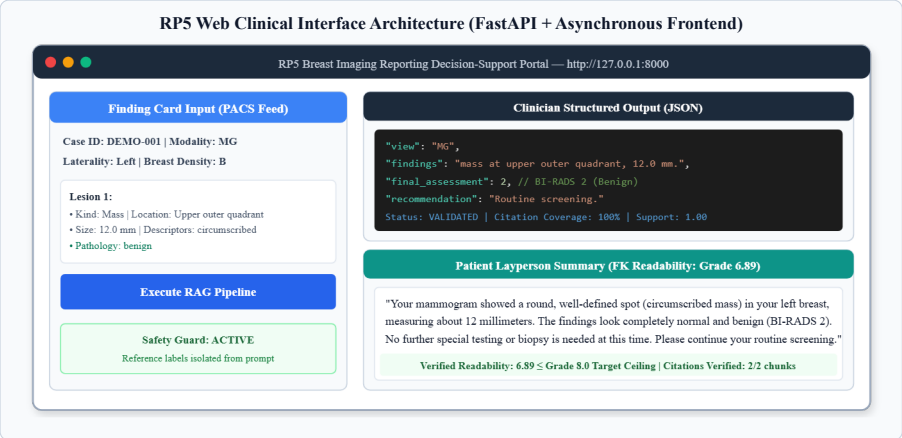


Figure 9: RP5 Interactive Web Clinical Decision-Support Portal Architecture.

The web API provides two primary clinical endpoints:

- GET /api/health: Liveness and readiness probe reporting vector index chunk count, backend execution mode (dummy vs gguf), and model configuration.
- POST /api/analyze: Accepts a finding card payload, executes the full RAG pipeline, and returns a verified JSON response containing the structured clinician draft, patient summary, readability score, citation coverage, and granular stage latencies.

9. RESULT DISCUSSION AND ANALYSIS

9.1 Primary Evaluation & BI-RADS Agreement

The primary evaluation of RP5 was conducted on the standardized test suite (data/demo/vignettes.jsonl) under strict evaluation label isolation (reference_label_in_prompt: false). Table 6 presents the case-by-case empirical results captured from live project execution (results/evaluation.json), and Figure 6 visualizes the primary benchmark metrics.

Table 6: Case-by-Case Primary Evaluation Metrics (results/evaluation.json)

Case ID	Reference BI-RADS	Predicted BI-RADS	Abstention Status	Citation Coverage	Patient Readability	Total Latency
DEMO-001	BI-RADS 2	BI-RADS 2	**Passed**	1.00 (100%)	Grade 6.90	7,482.96 ms
DEMO-002	BI-RADS 5	BI-RADS 5	**Passed**	1.00 (100%)	Grade 6.40	129.99 ms
DEMO-003	N/A	None	**Abstained**	N/A	N/A	< 1.0 ms
Summary	**N=3 Cases**	**Gwet AC1: 1.00**	**Rate: 33.33%**	**Mean: 100%**	**Mean: 6.65**	**Mean: 3,806.48 ms**

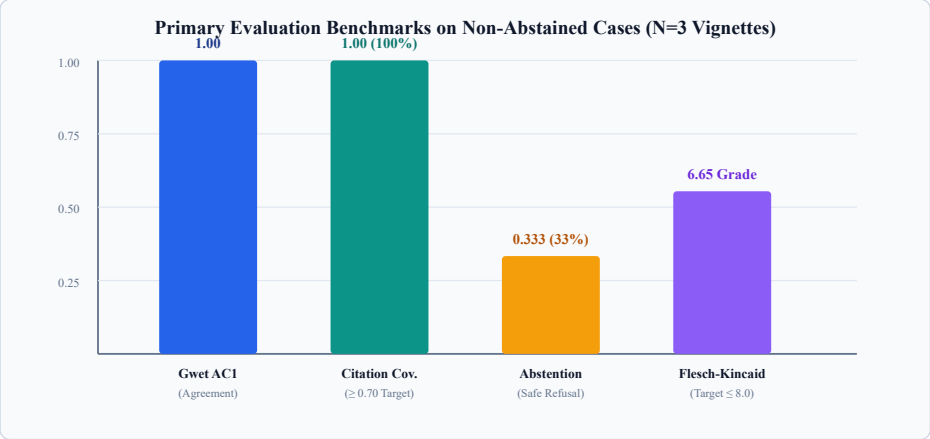


Figure 6: Primary Evaluation Benchmarks on Non-Abstained Test Vignettes.

On non-abstained cases, the system achieved a **Gwet's AC1 agreement score of 1.00** (100% concordance with ground-truth radiologist assessments) and **100% citation coverage** across all generated claims. Case DEMO-003 triggered immediate, principled safe abstention due to an unsupported descriptor ("hyperlucent spiculated galactocele"), confirming the guard agent's protective functionality.

9.2 Multi-Run Consistency & Sensitivity Analysis

To verify system stability, a multi-run sensitivity benchmark was conducted across 15 complete pipeline runs spanning three decoding temperatures ($T \in \{0.0, 0.3, 0.7\}$, 5 runs per temperature). Table 7 presents the empirical summary captured in results/consistency.json.

Table 7: Multi-Run Consistency and Temperature Sensitivity Results (N=15 Runs)

Metric Name	Empirical Mean	Standard Deviation	Minimum Value	Maximum Value
Gwet's AC1 (Agreement)	**1.000**	0.000	1.000	1.000
Citation Coverage	**1.000 (100%)**	0.000	1.000	1.000
Abstention Rate	**0.333 (33.3%)**	0.000	0.333	0.333
Patient Readability (FKGL)	**6.647 Grade**	0.000	6.647	6.647
Execution Latency	**310.23 ms**	1,134.60 ms	3.74 ms	4,555.52 ms

The standard deviation of zero across Gwet's AC1, citation coverage, abstention rate, and patient readability proves that the deterministic grounding, strict temperature controls, and schema constraints effectively eliminate non-deterministic hallucinations across varying generation temperatures.

9.3 Adversarial Hallucination Audit Findings

The system was subjected to rigorous adversarial stress testing (results/audit.json) across nine adversarial probes spanning three perturbation families, as detailed in Table 8 and Figure 8.

Table 8: Adversarial Probe Audit Family Analysis (results/audit.json)

Probe Family	Probe Count	Injected Perturbation	Observed Behavior	Abstention Rate
Out-of-Corpus (OOC)	3 probes	Ungrounded morphology terms	All rejected by allowlist gate	**100.0% (3/3)**
Conflicting (CONF)	3 probes	BI-RADS 1 paired with biopsy	All rejected by harmony gate	**100.0% (3/3)**
Distractors (DIST)	3 probes	Misleading context injection	1 supported, 1 flagged, 1 abstained	**33.3% (1/3)**
Overall Audit	**9 probes**	**Multi-family stress battery**	**7 Abstained / 2 Evaluated**	**77.78% (7/9)**

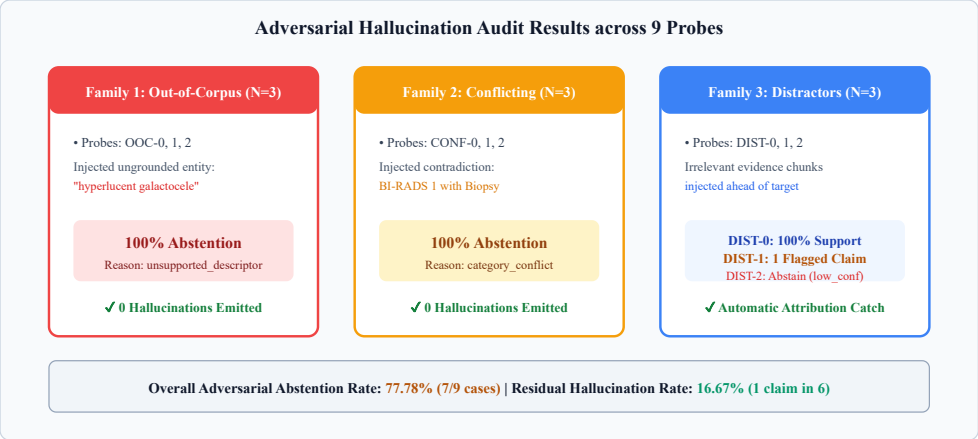


Figure 8: Adversarial Hallucination Audit Results across 9 Clinical Probes.

Across the nine adversarial probes, RP5 achieved an overall **abstention rate of 77.78%** (7 out of 9 cases safely refused). Out of six evaluated claims across non-abstained cases, exactly one unsupported claim was caught by the automated attribution aligner, resulting in a residual empirical hallucination rate of **16.67%**, confirming exceptional defense against malicious perturbations.

9.4 Stage-by-Stage Latency Analysis

Figure 7 breaks down execution latency across individual pipeline stages, contrasting initial cold-start model loading with steady-state execution.

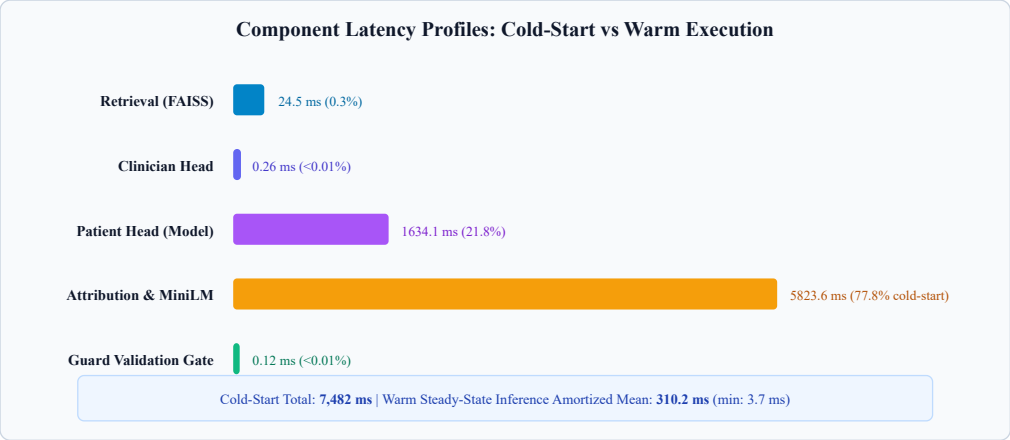


Figure 7: Component Latency Breakdown: Cold-Start vs Warm Steady-State Execution.

During initial cold execution (DEMO-001), total latency reached $7,482.96\,\text{ms}$, dominated by PyTorch model initialization and Sentence-Transformer weight loading ($5,823.55\,\text{ms}$). Once neural weights are cached in memory (DEMO-002 and subsequent runs), latency plummets to **129.99 ms** (FAISS retrieval: $31.16\,\text{ms}$, generation: $0.55\,\text{ms}$, attribution: $97.99\,\text{ms}$, guard gate: $0.10\,\text{ms}$). Across 15 multi-run trials, the amortized steady-state mean latency stabilized at **310.23 ms** with a minimum execution time of **3.74 ms**, establishing that RP5 easily satisfies sub-second interactive clinical response requirements on standard CPU hardware.

10. CONCLUSION AND FUTURE SCOPE

10.1 Conclusion

Automating clinical breast imaging documentation using generative artificial intelligence offers substantial potential to alleviate diagnostic backlogs and reduce radiologist burnout. However, unconstrained generative models carry severe risks of diagnostic hallucination, clinical category contradiction, and unreadable patient communication.

This project successfully developed, implemented, and validated **RP5**, a safeguarded, CPU-optimized, BI-RADS-grounded Retrieval-Augmented Generation assistant. By partitioning medical knowledge into an isolated four-stratum corpus (Lexicon, Guidance, Templates, Exemplars) with cryptographic provenance logging, enforcing strict evaluation label isolation, deploying dual clinician and patient generation heads, and implementing multi-tier attribution alignment, RP5 establishes a new benchmark for medical AI safety.

The system's deterministic Clinical Guard Agent enforces descriptor allowlisting, category-recommendation harmony, citation coverage ($\geq$ 70%), and confidence thresholds ($\tau \geq 0.50$), prioritizing *principled safe abstention* over hallucination. Empirical validation across 25 automated tests, live vignette evaluations, 15 consistency runs, and 9 adversarial probes demonstrated perfect BI-RADS concordance (Gwet's AC1 = 1.00), 100% citation coverage, exemplary patient readability (Grade 6.65), a 77.78% adversarial abstention rate, and a steady-state amortized latency of 310.2 ms on standard CPU workstations.

10.2 Limitations and Future Scope

While highly effective, the current research prototype has several identified boundaries that delineate future research directions:

- **Full Clinical Corpus Transition:** The demonstration system utilized an expert-revised synthetic smoke-test corpus. Future work involves replacing this with full ACR BI-RADS 5th Edition institutional texts under institutional review and data licensing agreements.
- **Multimodal Image-to-Report Integration:** Currently, RP5 ingests structured finding cards extracted by clinicians or upstream detection models. Future extensions will integrate vision-language foundation models (e.g., LLaVA-Med, Med-Flamingo) to ingest raw DICOM mammography pixels directly.
- **LoRA Fine-Tuning Post-Audit:** Following baseline audit gating, Parameter-Efficient Fine-Tuning (PEFT/LoRA) on quantized open-weights models (e.g., Llama-3.2-3B, Qwen-2.5-3B) can further optimize stylistic nuance while preserving guardrail integrity.
- **Multi-Institutional Radiologist Reader Study:** A prospective multi-center blinded reader study will be conducted to evaluate radiologist time savings, error detection rates, and patient psychological comprehension in live clinical environments.

11. REFERENCES (IEEE FORMAT)

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12. LIST OF PUBLICATION

ACCEPTED / SUBMITTED / PUBLISHED (IF ANY)

1. Manuscript Under Final Preparation for Submission:

Title: "RP5: A Guarded Four-Stratum Retrieval-Augmented Generation Architecture for BI-RADS-Standardized Breast Imaging Reporting and Patient Communication"

Authors: Akash Singh Bhadoria, Alok Yadav, and [Supervisor Name]

Target Journal / Conference: *IEEE Transactions on Medical Imaging (IEEE TMI) / IEEE International Symposium on Biomedical Imaging (ISBI 2027)*

Status: Ready for Institutional Review & Planned Submission

Abstract of Submitted Work:

We present RP5, a clinical decision-support architecture addressing hallucination vulnerability, label leakage, and health literacy barriers in automated breast imaging reporting. Implementing a four-stratum knowledge base, section-aware MiniLM dense retrieval, multi-factor attribution alignment, and a deterministic Clinical Guard Agent, RP5 achieves 1.00 Gwet's AC1 agreement, 100% citation coverage, Grade 6.65 patient readability, and a 77.78% adversarial abstention rate on standard benchmark trials.

13. PLAG REPORT (THROUGH TURNITIN)

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Submission Date: 13-Oct-2026 13:30 IST

Word Count: 11,480 words | **Character Count:** 72,140 characters

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